

Computer Vision: Final Project Proposal

Spring 2026

This document describes the two options available for the CV course final project. Please read both options carefully, choose one, and submit a project proposal (1–2 pages) by the deadline stated on Web Learning (网络学堂) .

1. Option 1: Efficient Label-Noise Detection in ImageNet-1K Training Set

ImageNet-1K is one of the most influential datasets in computer vision, yet its training set is known to contain a non-negligible amount of label noise. In this project, your goal is to **efficiently identify mislabeled samples** in the ImageNet-1K training split.

1.1. Evaluation Criteria

Your submission will be scored along three axes (you may emphasize one or balance all three):

1. **Quantity**: The total number of verified incorrect labels you discover.
2. **Accuracy**: The precision of your detections (i.e., what fraction of flagged samples are actually mislabeled). A long list of random guesses will score poorly.
3. **Resource Efficiency**: The computational cost, wall-clock time, and human effort required to obtain your results. Both effectiveness and resource consumption will be considered in your final score.

You are free to pursue any strategy, for example:

- **Minimal-resource**: Build a lightweight pipeline that finds a respectable number of errors with very little compute.
- **Maximal-coverage**: Scale up computation to uncover as many mislabeled images as possible.
- **Best efficiency ratio**: Maximize the number of verified errors per unit resource (e.g., per GPU-hour or per human-review hour).

1.2. Allowed Methods & Tools

The list below is illustrative rather than exhaustive; any relevant method is welcome.

- **Pre-trained models**: Use off-the-shelf classifiers trained on ImageNet-1K or larger datasets. Techniques such as confidence scoring, margin analysis, entropy, or ensemble disagreement are all valid.
- **Vision-Language Models (VLMs)**: You may query VLM APIs to verify or generate candidate labels. Be aware that API calls count toward your resource budget.
- **Traditional CV methods**: Compute image similarity between training and validation images. If two images are visually almost identical but carry different ground-truth labels, at least one is likely wrong.
- **Human review**: Manual verification is explicitly allowed and often necessary for final confirmation. However, label your time spent on curation carefully; human hours are part of the resource budget.

1.3. Budget & Reimbursement

- **VLM API costs** are **not** reimbursable.
- **GPU rental expenses** are reimbursable up to **400 CNY per student**.

1.4. Team Size

This option is to be completed individually (solo).

2. Option 2: Open-Ended Project

If you prefer to explore a topic of your own interest, you may propose a self-designed project. The topic must be related to computer vision or visual learning, but otherwise the scope is flexible.

2.1. Suggested Topics

The following examples are meant to inspire you; you are not restricted to this list.

- **Traditional CV:** Camera calibration from scratch, including chessboard corner detection, homography estimation, intrinsic / extrinsic parameter decomposition, and reprojection error evaluation.
- **CV Applications:** Build a real-time pipeline that uses a webcam to capture a game screen, detect in-game heads, and overlay an auto-aim reticle (for educational / demonstration purposes only).
- **Visual Neural Networks**
- **Vision-Language Models**
- **Generative Models**

2.2. Proposal Requirements

Because this option is open-ended, your proposal must be especially clear:

1. **Problem Statement:** What exactly are you trying to solve? Why is it interesting or useful?
2. **Related Work:** Cite 2–4 relevant papers or existing solutions.
3. **Methodology:** Describe the techniques, models, or algorithms you plan to use.
4. **Evaluation Plan:** How will you measure success? Define quantitative metrics and, if applicable, qualitative criteria.
5. **Timeline & Milestones:** Provide a week-by-week plan with concrete deliverables.

2.3. Budget & Reimbursement

- **VLM API costs** are **not** reimbursable.
- **GPU rental expenses** are reimbursable up to **400 CNY per student**.

2.4. Team Size

You may work alone or in a team of two (1–2 people).

3. Submission Instructions

1. **Proposal Deadline:** Submit a 1–2 page PDF proposal via Web Learning (网络学堂) by **May 17, 2026, 23:59:00 (UTC+8)**. Clearly state which option you have chosen.
2. **Final Deadline:** Submit your code, data (or links), and final report by **June 21, 2026, 23:59:00 (UTC+8)** via Web Learning (网络学堂) .
3. **Team Size:** Option 1 must be completed individually. Option 2 may be completed by 1–2 people. All code, analysis, and writing must be your own.